(1.) Course Title: INFPHD080 Systems of language processors: formal – language-theoretic models of multi-agent systems	**Credits: 6**
Course **Classification: Optional**	
Theoretical or Practical Nature of the Course, *'training character'*[13]: 90% theory (credit%)	
Type of Class: lecture and **number of hours**: **22** in the given semester, *(If the course is not (only) taught in Hungarian, then **the language**: **English**)* **Additional** (specific**) methods and characteristics**[2] used in delivering the knowledge *(if any)*:	
Assessment Method (Exam / Midterm Grade / **Other**[3]): **Additional** (specific) **methods**[4] applied in knowledge assessment *(if any)*:	
Course Placement in Curriculum (Semester Number): **2-3**	
Prerequisites (*if any*): **basic knowledge in formal language and automata theory and in distributed computing**	

Course Description: **A concise yet informative description of the knowledge** to be acquired.

Distributed computation models are of particular importance in artificial intelligence research, in the theory of nature-motivated, including biologically motivated, computation, and in the development of the mathematical foundations of distributed processing of natural languages.
The planned lecture series will describe the properties and behaviour of distributed computation models using formal language tools.
The lectures will cover the following topics:

Cooperating/Distributed Grammar Systems: syntactic modeling of the well-known table architecture from artificial intelligence research, where the cooperating agents are represented by formal grammars (language processors) and the common database (the table) they use for problem solving is represented by the language generated by the processors. Key models, cooperation strategies. Size and computational complexity.

Colonies: characterisation of the eemergent behaviour of communities of extremely simple, purely reactive agents using formal grammars. Key model classes.

Eco-grammar systems: formal-language-theoretic models of artificial life, describing dynamically changing, evolving and adaptive agents and the environment they share, evolve and interact with.

Networks of Language Processors:
Language processing networks: computational models based on massively parallel formal grammars. Language processors are located at the vertices of the (virtual) network, processing and communicating words, sets of words or multisets of words. Major classes, instances of on-demand and on-command communication. Biologically motivated models: test-tube systems known from DNA computation, networks of evolutionary processors, Watson Crick L systems. Networks of reaction systems.
Size, computational and communication complexity. Phenomena in networks. Novel measures of complexity.

2-5 Most Important Required and Recommended Literature

List of required and recommended literature (notes, textbooks) with bibliographic data (author, title, publication details, (possibly pages), ISBN).

- E. Csuhaj-Varjú, J. Dassow, J. Kelemen, Gh. Paun: Grammar Systems: A Grammatical Approach to Distribution and Cooperation. Gordon and Breach Science Publishers, Yverdon, 1994.
- J. Dassow, Gh. Paun, G. Rozenberg: Grammar Systems. In: Handbook of Formal Languages. (G. Rozenberg, A. Salomaa, eds.), Vol 2, Chapter 4, Springer, 1997.
- E. Csuhaj-Varjú: Grammar Systems. In: Formal Languages and Applications. Studies in Fuzziness and Soft Computing 148. (C. Martín-Vide, V. Mitrana, Gh. Paun, eds.), Springer, Berlin, 2004, 275-310.
- Gh. Paun, G. Rozenberg, A. Salomaa: DNA Computing - New Computing Paradigms, Springer Verlag, Berlin, 1998.
- Multiagent Systems. A Modern Approach to Distributed Artificial Intelligence. (G. Weiss, ed.), The MIT Press, Cambridge, Massachusetts, 1999.
- J. Castellanos, C. Martín-Vide, V. Mitrana, *et al.* Networks of evolutionary processors. *Acta Informatica* **39**, 517–529 (2003)
- A. Ehrenfeucht, G. Rozenberg: Reaction Systems. Fundamenta Informaticae **75** (1-4) 263-280, (2007)
- Recent literature on „Networks of Evolutionary Processors" and „Reaction Systems".

List of Prescribed Professional Competencies and Competency Elements

(knowledge, skills, etc., KKK Section 8) that the course significantly contributes to developing (highlight the most important ones).

a) Knowledge

- Has a research-level understanding of the general and specific characteristics, major directions, and precisely developed boundaries of the subject area within the discipline, as well as the agreed and debated interrelations.
- Creatively comprehends the interrelations, theories, and conceptual frameworks and terminologies of the given field of study.1
- Possesses research methodology knowledge necessary for independent research in the given discipline/field.

b) Skills

- Capable of creative analysis of the field, synthesizing comprehensive and specific interrelations with a new perspective, and conducting appropriate evaluative and critical activities.
- Able to apply and further develop the specialized learning and problem-solving methods of the field.
- Can creatively elaborate on the novel, previously unknown practical implications of theoretical issues.

[1] **Section 108**, Article 37 of the National Curriculum Act (Nftv.) 37. lesson: a session (lecture, seminar, exercise, consultation) requiring the personal involvement of the instructor to fulfil the academic requirements specified in the curriculum, the duration of which is at least forty-five minutes and not more than sixty minutes.

[2] e.g. case studies, role play, thematic presentations, etc.

[3] e.g. continuous debriefing, mid-year report

[4] e.g. request for case studies, thematic papers, essays, business, organisational plans, etc.

- Capable of planning and executing new projects, conducting research in the field, and developing new techniques and approaches.
- Able to identify unforeseen professional problems and uncover the detailed theoretical and practical background necessary for their resolution at a research level.
- Can construct and communicate new relations relevant to the field, as well as comprehensive interrelations significant for personal and communal existence.

c) Attitudes

- Represents and further develops the relations that contribute to the process of human self-creation based on the field's characteristics.
- Possesses an interest and learning ability that enable identifying and solving currently unpredictable research problems in the field.
- Demonstrates a firm professional commitment, a continuous dedication to exploring new approaches, and an acceptance of the necessity for persistent effort.

d) Autonomy and Responsibility

- Builds and initiates new knowledge areas with creative independence and proposes new practical solutions.
- Participates in formulating theoretical and practical questions with a leadership role and high-level cooperation.
- Engages as an equal discussion partner with professionals in the field.
- Takes responsibility for raising and addressing new ethical questions concerning theoretical and practical aspects of the profession.

Course responsible (name, position, academic degree): **Erzsébet Csuhaj-Varjú, prof. emerita, DSc**

Instructor(s) involved in teaching the course (if any) (name, position, academic degree):
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